UBT619: PHARMACEUTICAL TECHNOLOGY

L	T	P	Cr
3	0	0	3.0

Course Objective: The objective of this course is to make students understand the basic concepts involved in pharmaceutical industry. The course will give knowledge about new drug development and approval process, ADMET of drugs, about the manufacturing and quality control of conventional, new type of dosage forms and biotechnology derived pharmaceuticals.

Syllabus

Introduction to drugs and pharmacy: An overview and history of pharmaceutical industry. The business and the future of Biopharmaceuticals. Drug regulation and control. Scope and applications of biotechnology in pharmacy. New drug development and approval process: Strategies for new drug discovery, finding a lead compound, combinatorial approaches to new drug discovery, pre-clinical and clinical trials

Drug pharmacokinetics & pharmacodynamics: Routes of drug administration, membrane transport of drugs, absorption, distribution, metabolism and excretion of drugs. Factors modifying drug action, mechanism of drug action on human beings, receptor theory of drug action, pharmacogenomics, adverse effects of drugs and toxicology, Drug interactions

Pharmaceutical manufacturing: Drug dosage forms and their classification. Sterile dosage forms- parenteral and biologics, novel dosage forms and targeted drug delivery systems. Current good manufacturing practices and issues. Quality control of pharmaceutical products as per pharmacopoeia. Stability studies, Method validation

Biotechnology derived pharmaceuticals: Production of pharmaceuticals by genetically engineered cells- hormones and vaccines. Drug regulatory requirement of India, Drug regulatory and accrediting agencies of the world (USFDA, TGA, ICH, WHO, ISO etc.). Overview of registration process of Indian drug product in overseas market

Course Learning Objectives (CLO)
The students will be able to:

1. explain the regulatory aspects and various steps of new drug discovery process.
2. explain the concept of pharmacodynamics and pharmacokinetics.
3. apply the knowledge of pharmaceutical manufacturing in the production of biopharmaceuticals like antibiotics, vaccines, proteins and hormones.
4. carry out the quality control procedures in the production of various biopharmaceuticals.

Text Books

1. Allen, L.V., Popovich, N.G. and Ansel, H.C., Ansel's Pharmaceutical Dosage Forms and Drug Delivery Systems, Lippincott Williams and Wilkins (2005).
2. Walsh, G., Biopharmaceuticals: Biochemistry and Biotechnology, Wiley (1998).

Reference Books

1. Gennaro, A.R., Remington: The Science and Practice of Pharmacy. Lippincott Williams and Wilkins (2005).
2. Tripathi, K.D., Essentials of Medical Pharmacology, Jaypee Brothers Medical Publishers (2008).

Evaluation Scheme:

Sr.No.	Evaluation Elements	Weightage (%)
1.	MST	25-35
2.	EST	35-45
3.	Sessionals (May include assignments/quizzes)	30